Jerry Wu

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Experience

Software Engineering Intern May 2026 – Aug. 2026
Carta *Waterloo, ON (Hybrid)*

- Automated nightly sales-demo environment refreshes using PostgreSQL snapshots in S3, KMS, and Kubernetes CronJobs deployed with Helm, saving **2.5 hours per week** in manual seeding while preserving record IDs and stable demo URLs
- Restored access-control state after demo refreshes across **50+ services** using an idempotent Spring Boot/Quartz replay job; provisioned least-privilege S3/KMS permissions with Terraform
- **Reduced median request latency by 10%** across 8 Django API endpoints by offloading non-critical persistence, event processing, and downstream API calls to asynchronous Kafka consumers
- Cut **CircleCI pipeline runtime by 33%**, from 18 to 12 minutes, by parallelizing Java builds, decoupling Error Prone checks from the critical path, and tuning job concurrency across 5 pipeline stages

Software Engineering Intern, Fixed Income May 2025 – Sept. 2025
PengYang Asset Management *Beijing, China*

- Reduced portfolio simulation and backtesting runtime to **under 5 seconds** using vectorized pandas operations and multithreading, working with **10 years of hourly market data** across thousands of Chinese equities
- Built Python/pandas ETL pipelines to process fund positions and performance data into portfolio-level metrics; implemented Black–Litterman asset allocation with confidence-adjusted analyst views

Technical Skills

Languages: C++, C, Python, Java, TypeScript, SQL
Systems & Performance: LLVM/Clang, CUDA, Linux, CMake, Multithreading, Memory Management
Backend & Data: Django, Spring Boot, FastAPI, Kafka, PostgreSQL
Cloud & Infrastructure: AWS (S3, KMS), Kubernetes, Docker, Helm, Terraform, CircleCI
Frameworks & ML: React, Next.js, PyTorch, NumPy, pandas

Open Source

LLVM / Clang ↗ | *C++, Compiler Frontend, AST* Oct. 2026

- Developed a Clang semantic-analysis fix for invalid C++ `consteval` calls in lambda init-captures while preserving closure types; credited for parallel work in **merged PR #234793** ↗
- Added C++20/23 regression tests for mixed captures, `consteval` constructors/functions, and source locations; verified the merged fix across **14 focused test configurations**

Projects

LLM Inference Engine ↗ | *C++20, CUDA, NVFP4/FP8, CMake*

- Built a native C++20/CUDA inference runtime for **Qwen3.6-27B-NVFP4**, implementing hybrid Gated-DeltaNet/full-attention, chunked prefill, and custom NVFP4/FP8 linear kernels
- Ran the 27B model on a **16 GB RTX 5080** by keeping **44 of 64 layers** resident in VRAM and streaming the remaining 5.28 GB from pinned host memory with double-buffered asynchronous prefetch
- Validated outputs from all 64 layers against a NumPy reference implementation and achieved **6.48 tokens/s** during greedy decoding

TruthTone++ ↗ | *Python, PyTorch, FastAPI, Next.js*

- Won **Best Audio AI Hack** at CxC 2026, a University of Waterloo hackathon with **350+ participating teams**; developed an audio deepfake classifier using Mel spectrograms and an ImageNet-pretrained **ResNet-34**
- Adapted ResNet-34 for single-channel spectrograms and built a Next.js/FastAPI app supporting microphone recordings, audio uploads, and per-second prediction heatmaps; published model checkpoints on Hugging Face

Education

University of Waterloo Waterloo, ON
Bachelor of Computer Science *Sept. 2024 – Expected 2028*

- Coursework: Data Structures & Algorithms, OOP, Convex Optimization, Compilers, Linear Algebra